

## MENOMINEE HARBOR, MI, USA

WMO: 997785

Lat: 45.0960N

Lon: 87.5900W

Elev: 178

StdP: 99.20

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |            |            |            |            | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|------------|------------|------------|------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |            | 99%        |            |            | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB       | DP         | HR         | MCDB       | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-21.0 | (c)<br>-18.6 | (d)<br>N/A                    | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>12.8           | (k)<br>-6.2 | (l)<br>11.7 | (m)<br>-6.0 | (n)<br>4.2            | (o)<br>270 | (p)<br>0.617 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |            |            |            |            |            | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|------------|------------|------------|------------|------------|----------------------|------------|
|               |                        | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |            | 1%         |            | 2%         |            |                      |            |
|               |                        | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB       | WB         | MCDB       | WB         | MCDB       | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>7.0             | (c)<br>28.6     | (d)<br>N/A | (e)<br>26.9 | (f)<br>N/A | (g)<br>25.4 | (h)<br>N/A | (i)<br>N/A          | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>4.7           | (p)<br>180 |

| Dehumidification DP/MCDB and HR |            |            |            |            |            |            |            |            | Enthalpy/MCDB |            |            |            |            |            | Extreme<br>Max WB |
|---------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|---------------|------------|------------|------------|------------|------------|-------------------|
| 0.4%                            |            |            | 1%         |            |            | 2%         |            |            | 0.4%          |            | 1%         |            | 2%         |            |                   |
| DP                              | HR         | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth          | MCDB       | Enth       | MCDB       | Enth       | MCDB       |                   |
| (a)<br>N/A                      | (b)<br>N/A | (c)<br>N/A | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A    | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A | (p)<br>N/A        |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |  |  |
|-------------------|----------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|--|--|
|                   |          |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |  |  |
| 1%                | 2.5%     | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |  |  |
| (a) 11.9          | (b) 10.5 | (c) 9.0 | (e) -23.7                  | (f) 32.1 | (g) 3.4            | (h) 1.7 | (i) -26.2                                          | (j) 33.3 | (k) -28.2  | (l) 34.2 | (m) -30.1  | (n) 35.2 | (o) -32.6  | (p) 36.4 |  |  |
|                   |          |         | DB                         | N/A      | N/A                | N/A     | N/A                                                | N/A      | N/A        | N/A      | N/A        | N/A      | N/A        | N/A      |  |  |
|                   |          |         | WB                         | N/A      | N/A                | N/A     | N/A                                                | N/A      | N/A        | N/A      | N/A        | N/A      | N/A        | N/A      |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 7.0    | -7.7  | -7.4  | -1.7  | 4.8   | 11.7  | 17.6  | 21.1  | 20.3  | 15.9  | 9.7   | 2.5   | -3.5  |
|                                                                               | DBStd       | 11.09  | 6.18  | 5.80  | 5.83  | 3.97  | 4.12  | 3.50  | 3.07  | 2.59  | 3.88  | 4.14  | 4.96  | 5.15  |
|                                                                               | HDD10.0     | 2296   | 549   | 487   | 363   | 164   | 28    | 0     | 0     | 0     | 2     | 56    | 228   | 418   |
|                                                                               | HDD18.3     | 4353   | 808   | 721   | 621   | 407   | 208   | 54    | 10    | 10    | 94    | 269   | 474   | 677   |
|                                                                               | CDD10.0     | 1208   | 0     | 0     | 1     | 7     | 81    | 228   | 343   | 319   | 178   | 47    | 3     | 0     |
|                                                                               | CDD18.3     | 223    | 0     | 0     | 0     | 0     | 3     | 32    | 95    | 71    | 20    | 2     | 0     | 0     |
|                                                                               | CDH23.3     | 903    | 0     | 0     | 0     | 2     | 16    | 128   | 455   | 236   | 60    | 6     | 0     | 0     |
|                                                                               | CDH26.7     | 164    | 0     | 0     | 0     | 1     | 2     | 18    | 101   | 34    | 7     | 1     | 0     | 0     |
| Wind                                                                          | WSAvg       | 4.5    | 4.6   | 4.8   | 4.3   | 5.0   | 4.8   | 4.2   | 3.9   | 3.8   | 4.4   | 4.9   | 4.9   | 4.8   |
| Precipitation                                                                 | PrecAvg     | 763    | 38    | 26    | 43    | 67    | 88    | 98    | 90    | 77    | 81    | 75    | 50    | 40    |
|                                                                               | PrecMax     | 958    | 138   | 56    | 101   | 119   | 144   | 223   | 164   | 143   | 157   | 181   | 146   | 89    |
|                                                                               | PrecMin     | 583    | 10    | 4     | 9     | 18    | 35    | 33    | 29    | 13    | 32    | 11    | 6     | 6     |
|                                                                               | PrecStd     | 91     | 28    | 12    | 23    | 30    | 31    | 57    | 39    | 32    | 35    | 43    | 33    | 23    |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 6.4   | 6.2   | 15.6  | 18.7  | 25.6  | 29.2  | 32.0  | 29.8  | 27.5  | 23.7  | 15.5  | 7.6   |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 2%          | DB     | 4.2   | 3.6   | 11.3  | 14.8  | 22.4  | 26.7  | 29.1  | 27.6  | 25.0  | 19.9  | 12.6  | 5.4   |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 5%          | DB     | 2.5   | 1.9   | 8.0   | 12.1  | 20.1  | 24.9  | 27.4  | 25.7  | 23.1  | 17.3  | 10.7  | 4.2   |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 10%         | DB     | 0.9   | 0.6   | 5.6   | 10.1  | 18.2  | 22.9  | 25.8  | 24.5  | 21.6  | 15.8  | 9.1   | 3.0   |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 6.7   | 7.5   | 7.7   | 6.1   | 6.7   | 6.4   | 7.0   | 6.6   | 6.9   | 5.7   | 5.3   | 5.5   |
|                                                                               |             | MCDBR  | 7.0   | 7.5   | 9.7   | 9.8   | 10.2  | 9.3   | 8.6   | 8.0   | 8.5   | 7.4   | 6.2   | 5.5   |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        |        | 0.261 | 0.290 | 0.329 | 0.360 | 0.392 | 0.395 | 0.386 | 0.388 | 0.364 | 0.330 | 0.300 | 0.271 |
|                                                                               | taud        |        | 2.463 | 2.363 | 2.371 | 2.357 | 2.323 | 2.351 | 2.393 | 2.381 | 2.442 | 2.527 | 2.529 | 2.494 |
|                                                                               | Ebn at Noon |        | 873   | 898   | 904   | 901   | 877   | 874   | 877   | 863   | 857   | 845   | 816   | 822   |
|                                                                               | Edh at Noon |        | 68    | 91    | 105   | 116   | 124   | 121   | 115   | 112   | 97    | 78    | 65    | 60    |
| All-Sky Solar<br>Radiation                                                    | RadAvg      |        | 1.42  | 2.42  | 3.62  | 4.56  | 5.54  | 5.99  | 6.16  | 5.23  | 4.05  | 2.49  | 1.48  | 1.09  |
|                                                                               | RadStd      |        | 0.15  | 0.26  | 0.33  | 0.43  | 0.45  | 0.46  | 0.26  | 0.35  | 0.30  | 0.26  | 0.14  | 0.11  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (49 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page